

TEZPUR UNIVERSITY
Major II, Autumn 2015

C0 504 Natural Language Processing

Marks : 40

Time : 60 minutes

The figures in the right-hand margin indicate marks for the individual questions.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. Why can digital storage of speech be expected to be more reliable than analog storage? 3
2. What are *words* and *morphemes*? Give examples of *irregular* inflection. 3 + 2
3. How can a lexicon be useful in NLP? 2
4. What are *terminal*, *non-terminal* and start symbols in a context free grammar for a *natural language*? 3
5. (a) What are the additional challenges in parsing a natural language expression using a context free grammar, compared to that of parsing by a compiler? 3
(b) Give an example of left recursion? How does it effect parsing? 2 + 2
(c) What is the *predictor* action in Earley's parser? 3
6. What will be the results of the following unifications- 2 + 2

$$\begin{aligned}
 & \text{(i)} \quad \left[\begin{array}{cc} \text{AGREEMENT} & \boxed{1} \left[\begin{array}{cc} \text{NUMBER} & \text{SG} \\ \text{PERSON} & 3 \end{array} \right] \\ \text{SUBJECT} & \left[\text{AGREEMENT} \quad \boxed{1} \right] \end{array} \right] \\
 & \sqcup \left[\begin{array}{cc} \text{SUBJECT} & \left[\text{AGREEMENT} \quad \left[\begin{array}{cc} \text{PERSON} & 3 \\ \text{NUMBER} & \text{SG} \end{array} \right] \right] \end{array} \right] \\
 & \text{(ii)} \quad \left[\begin{array}{cc} \text{AGREEMENT} & \boxed{1} \left[\begin{array}{cc} \text{NUMBER} & \text{SG} \\ \text{PERSON} & 3 \end{array} \right] \\ \text{SUBJECT} & \left[\text{AGREEMENT} \quad \boxed{1} \right] \end{array} \right] \\
 & \sqcup \left[\begin{array}{cc} \text{AGREEMENT} & \left[\begin{array}{cc} \text{NUMBER} & \text{SG} \\ \text{PERSON} & 3 \end{array} \right] \\ \text{SUBJECT} & \left[\text{AGREEMENT} \quad \left[\begin{array}{cc} \text{NUMBER} & \text{PL} \\ \text{PERSON} & 3 \end{array} \right] \right] \end{array} \right]
 \end{aligned}$$

7. How is a *probabilistic context free grammar (PCFG)* used in parsing? How can the probabilities in a PCFG obtained? 2 + 2
8. Give an example to show how *dependency parsing* helps parsing of relatively free word order language expressions. 2
9. Give example(s) of *ambiguity* and *vagueness* of natural language expressions. How should *ambiguity* and *vagueness* be considered in meaning representation in an NLP system? 2 + 2
10. Represent the folliwing expression using *First Order Predicate Calculus (FOPC)*- 3
TU students have good communication skills.